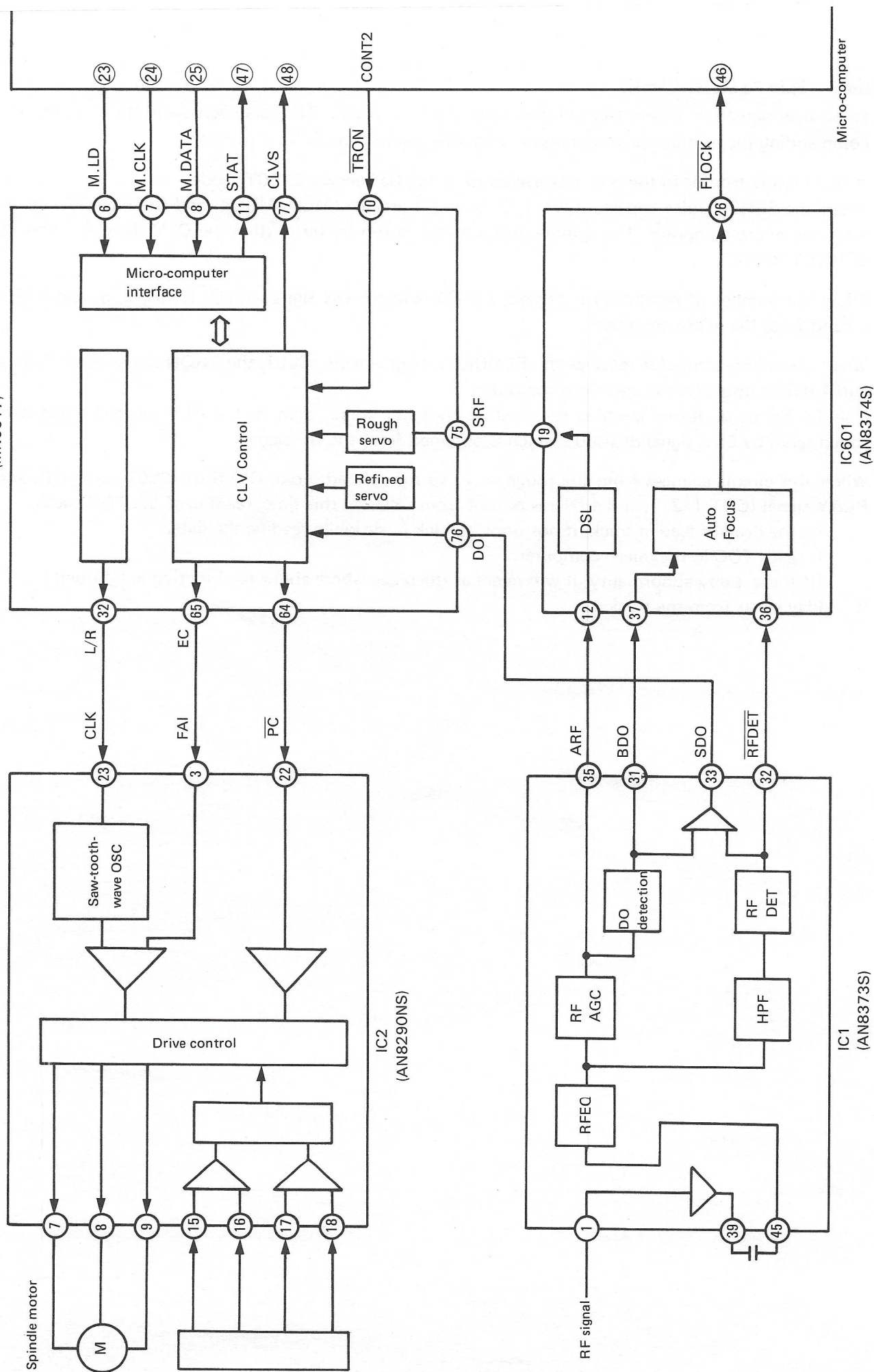


Fig. 4-4 Spindle Motor CLF-Control



A4-3 Spindle Motor Circuit

When disk detection ended, the spindle motor starts to rotate. The focus servo and the traverse servo are commanding the spindle motor operation; and after reading the TOC, play starts.

If there is any trouble in focus or traverse servo, it will become EJECT mode.

When the SENSE pulse appears, CNT 1, 2, 3, and 4 are (H). MLD, MCLK, and the MDATA are output from the micro-computer. The spindle motor starts to run by outputting the CLV data at pin 64 PC (L) of IC501 to IC2.

When the number of revolution increased, LD ON and AF ON signals (CNT 1, (L), 2, 3, and 4 (H)) are output from the micro-computer.

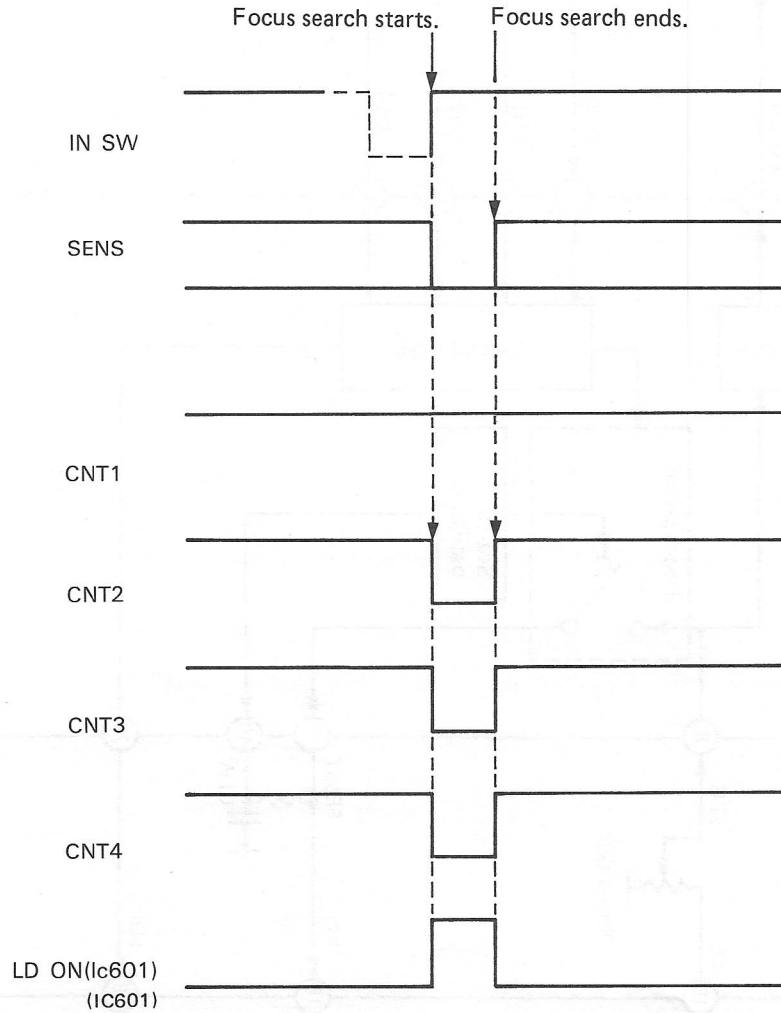
When the micro-computer receives the FLOCK (L) signal from IC601, the TRON signal (CNT 1, 2 (L), 3, and 4 (H)) is output from the micro-computer.

The TRON signal forms tracking servo while applying brake, then to the CLV control along with the rough servo by SRF signal of IC601 which is obtained from the RF signal.

When this circuit changes from the rough servo to the refined servo. CLVS of IC501 turns (H); and the PLAY signal (CNT 1, 2, 3, and 4 (H)) is output from CPU. At this time, reading of the TOC starts.

1. At the head of lead-in track, it positions by kick pulse while reading the data.
2. It reads TOC in the micro-computer.
(If there is any abnormality, it will reject at the place where abnormal function is affirmed.)
3. Play starts from the track 1.

(2) Focus Search Timing Chart



(1 "H", 0 "L")

HEX No.	Mode	Logic input				Logic control signal output							Remarks
		CNT1	CNT2	CNT3	CNT4	LD ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE	
0	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	Off track	Play mode
3	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	Off track	Brake mode
4	KICK CROSS	0	1	0	0	ON	ON	OFF	ON	OFF	OFF	CROSS	Kick mode
5	KICK REV	0	1	0	1	ON	ON	OFF	ON	OFF	ON (REV)	Shaping CROSS	
6	KICK FWD	0	1	1	0	ON	ON	OFF	ON	OFF	ON (FWD)	Shaping CROSS	
7	ACCESS	0	1	1	1	ON	ON	OFF	OFF	OFF	OFF	Shaping CROSS	Coarse feed
8	DISK DETECT	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S latch	Disk detection (Search Oscillation)
F	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaping CROSS	Forced return

Control Logic Table

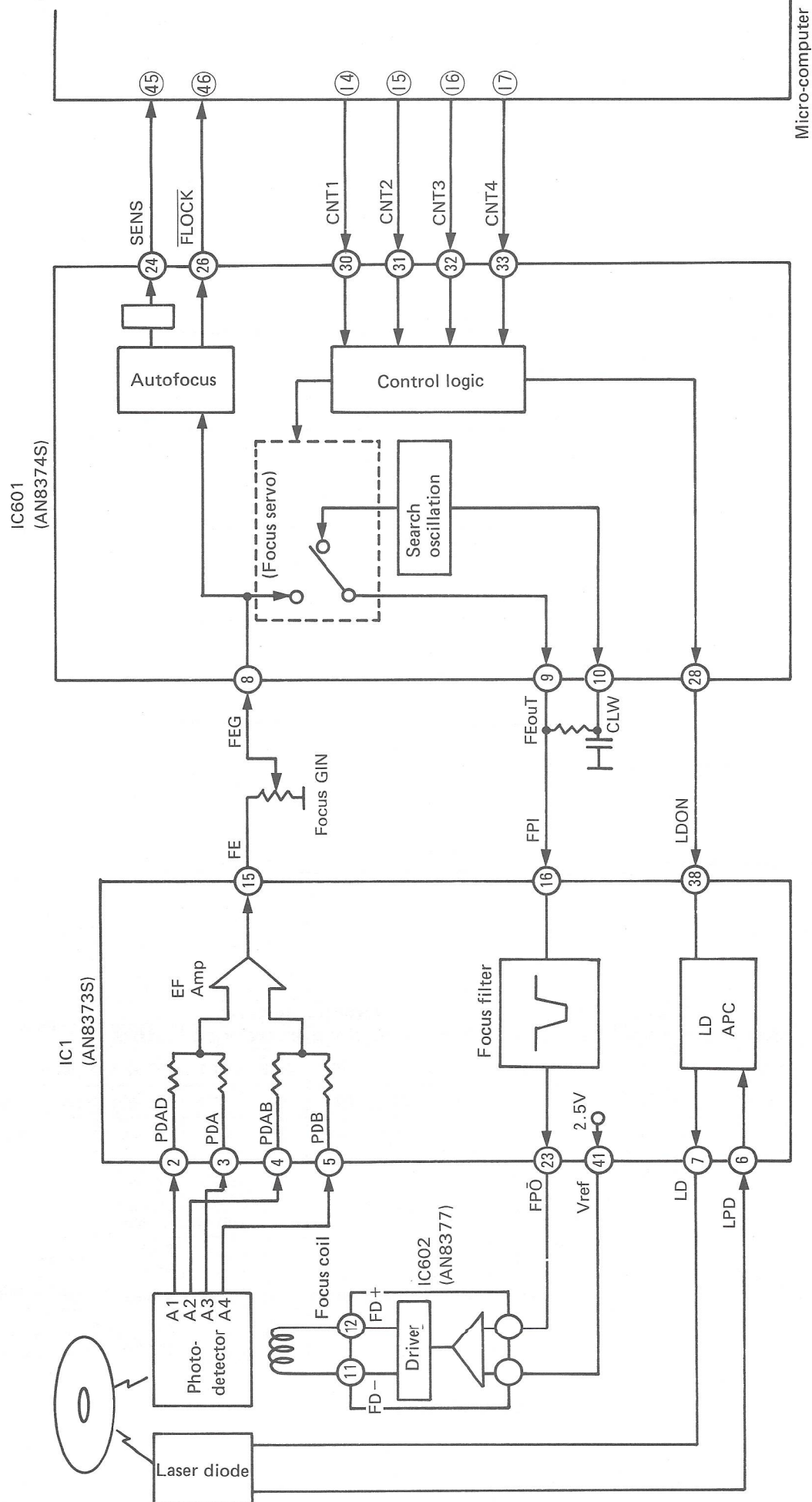


Fig. 4-3 Focus Control Circuit (Detecting Disk by Focus Search)

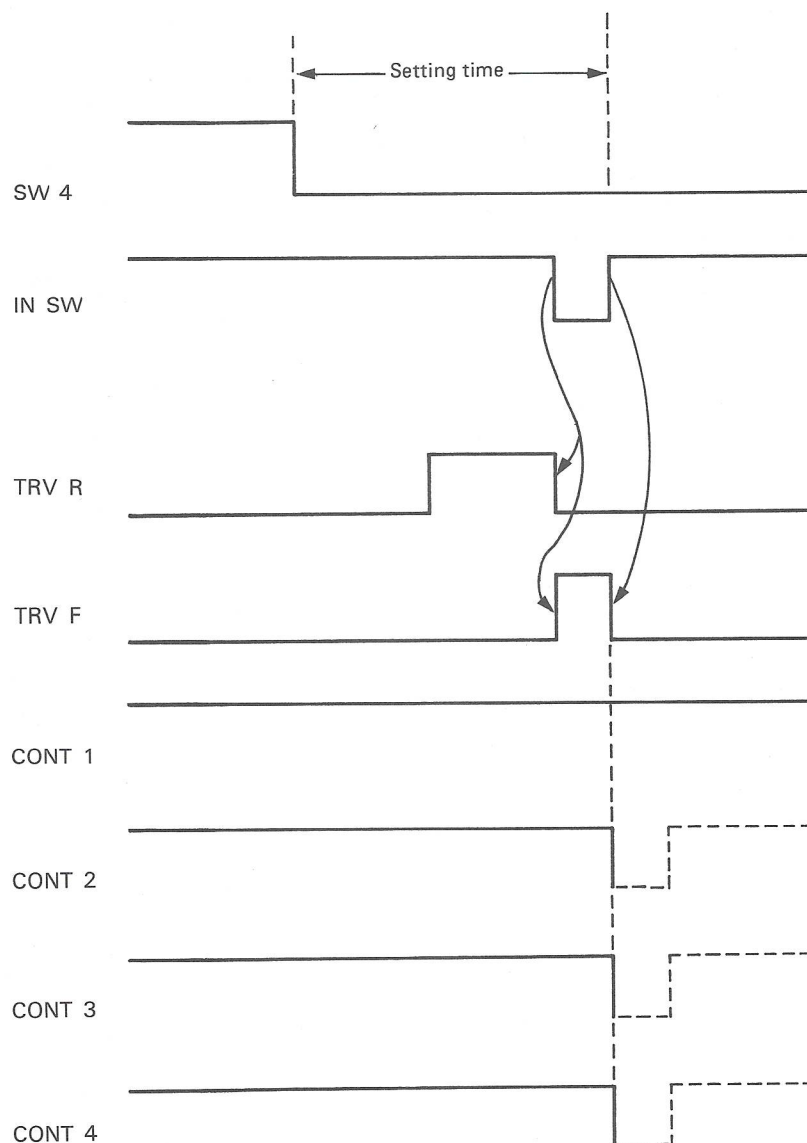
When the optical pickup starting position is set, this circuit makes focus search and detects the disk. The function is finished by the SENSE pulse (a disk exists) from IC601 pin 24.

By the riser edge from (L) to (H) of IN SW, disk detecting instruction (CNT 1 (H), 2, 3 and 4 (L)) is output from the micro-computer. Following this instruction IC601 instructs search oscillation and laser emission.

The search oscillation (triangular wave) appears at pin 10, along with FE out at pin 9, feed to IC1 pin 16. The output at pin 23, from the FPO terminal passes through the filter of IC1, drives the focus coil in IC602. As a result, the optical pickup moves up and down.

Meantime, the laser emission instruction LD ON at pin 28 of IC601 joins the LD APC circuit at pin 31 of IC1 LD ON. The laser power supplies at pin 7 of IC1, the laser emits.

The focus control circuit detects this state with the photo-detector and joins the FE signal at pin 15 of IC1 into IC601. And it detects the "S-curve" when the proper disk exists, and search oscillation as well as laser emission are normal. The detecting operation ends when (L) → (H) pulse from the SENSE applied to the micro-computer.



Traverse Controlling Logic

	TRV R	TRV F
OPEN	0	0
FORWARD	0	1
REVERSE	1	0
STOP	1	1

Fig. 4-2 Traverse Timing Chart (At setting of optical pickup starting position)

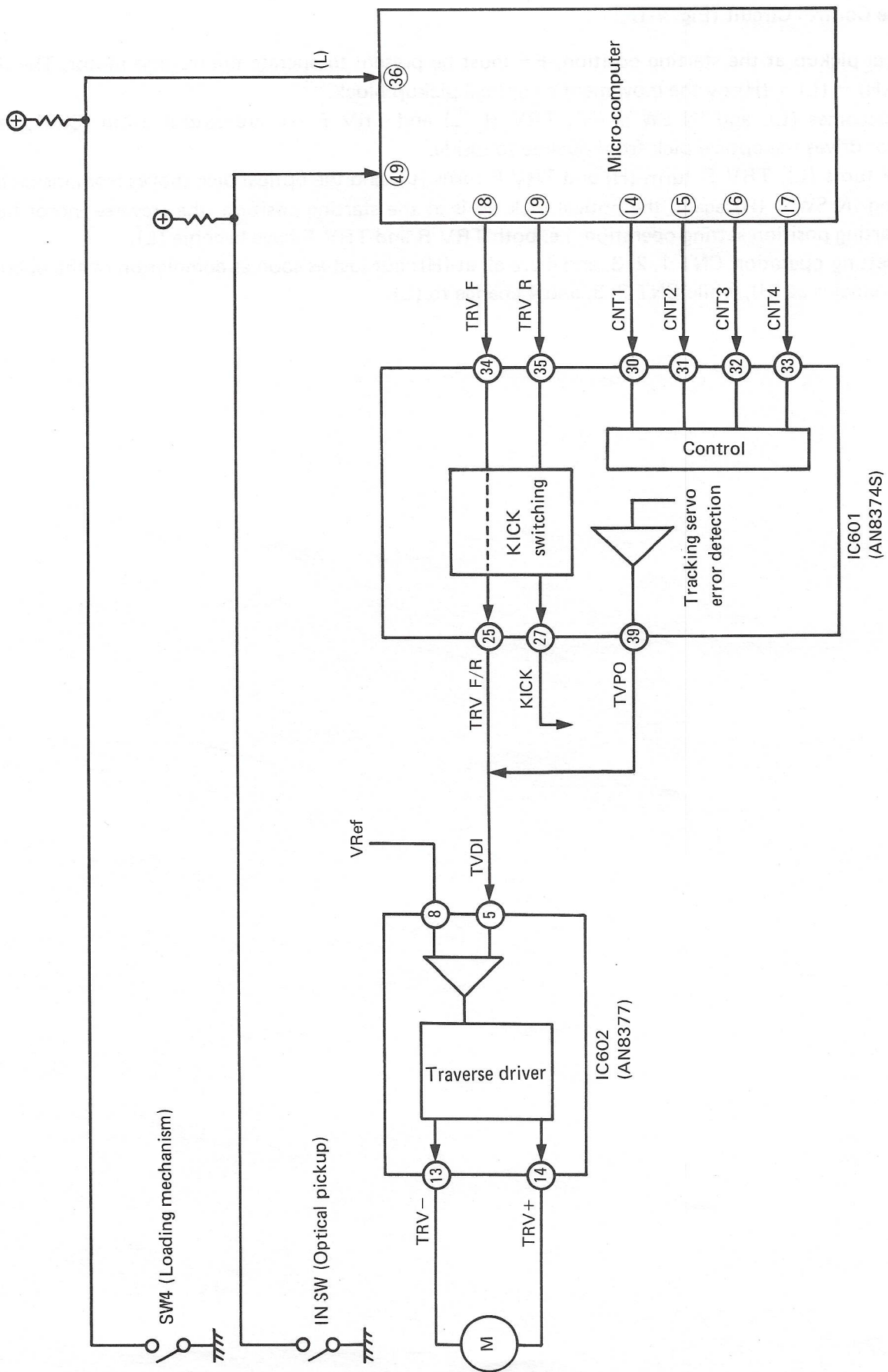


Fig. 4-1 Traverse Control Circuit (Optional pickup starting position setting)

4. CIRCUIT OPERATIONS (TYPE A)

4-1 Traverse Control Circuit (Fig. 4-1)

To the optical pickup at the starting position, RF must be present to operate the traverse motor. The IN SW changes (H) → (L) → (H) by the movement of optical pickup block.

When SW4 becomes (L), and IN SW is (H). TRV R (L) and TRV F (H) are output from IC601, the traverse motor drives the optical pick from outside to inside.

When IN SW turns (L), TRV R turns (H) and TRV F turns (L), and the optical pick moves from inside to outside. When IN SW is (H) again, the optical pick up is at the starting position, the traverse motor has ended the starting position setting operation, i.e. both TRV R and TRV F have become (L).

During the setting operation CNT 1, 2, 3, and 4 are all at (H); but just as soon as completion of the operation, CNT 1 remains at (H), while CNT 2, 3, and 4 change to (L).

(1) Sensor input

During disk loading or ejection operation, the microcomputer controls the loading motor and solenoid with four sensor inputs.

When a disk is set, the sensor information input to the microcomputer is as follows.

SW3	(P93)	0V
SW2	(P92)	0V
Door SW	(P83)	5V
Drop detection SW	(P91)	5V

Loading starts when door SW or SW2 switches on the following:

- Power supply
- Solenoid
- Loading motor

Eject starts when EJECT SW is pressed.

(2) Solenoid control

The solenoid is controlled by PA ϕ and PA1 outputs.

PA1 is for starting the solenoid.

PA ϕ is for holding the solenoid.

The solenoid operates when the disk goes in or out. (0V during operation)

(3) Loading motor control

There are three loading motor control modes.

- Forward operation for loading
- Reverse operation for eject
- Stop

Pin \ Mode		Loading	Eject	Stop
IC10	P ⑫	5V	0	5V
	P ⑬	0	5V	5V
IC11	P ⑤ ⑨	0	10V	0
	P ③ ⑦	10V	0	0

IC11 is a complementary transistor driver, whose equivalent circuit is shown in Figure 3-3.

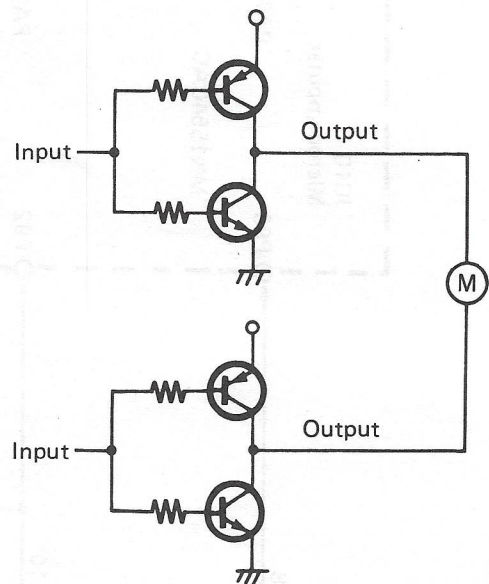


Fig. 3-3 Loading Motor Driver Circuit

C3-4 Disk Loading Circuit

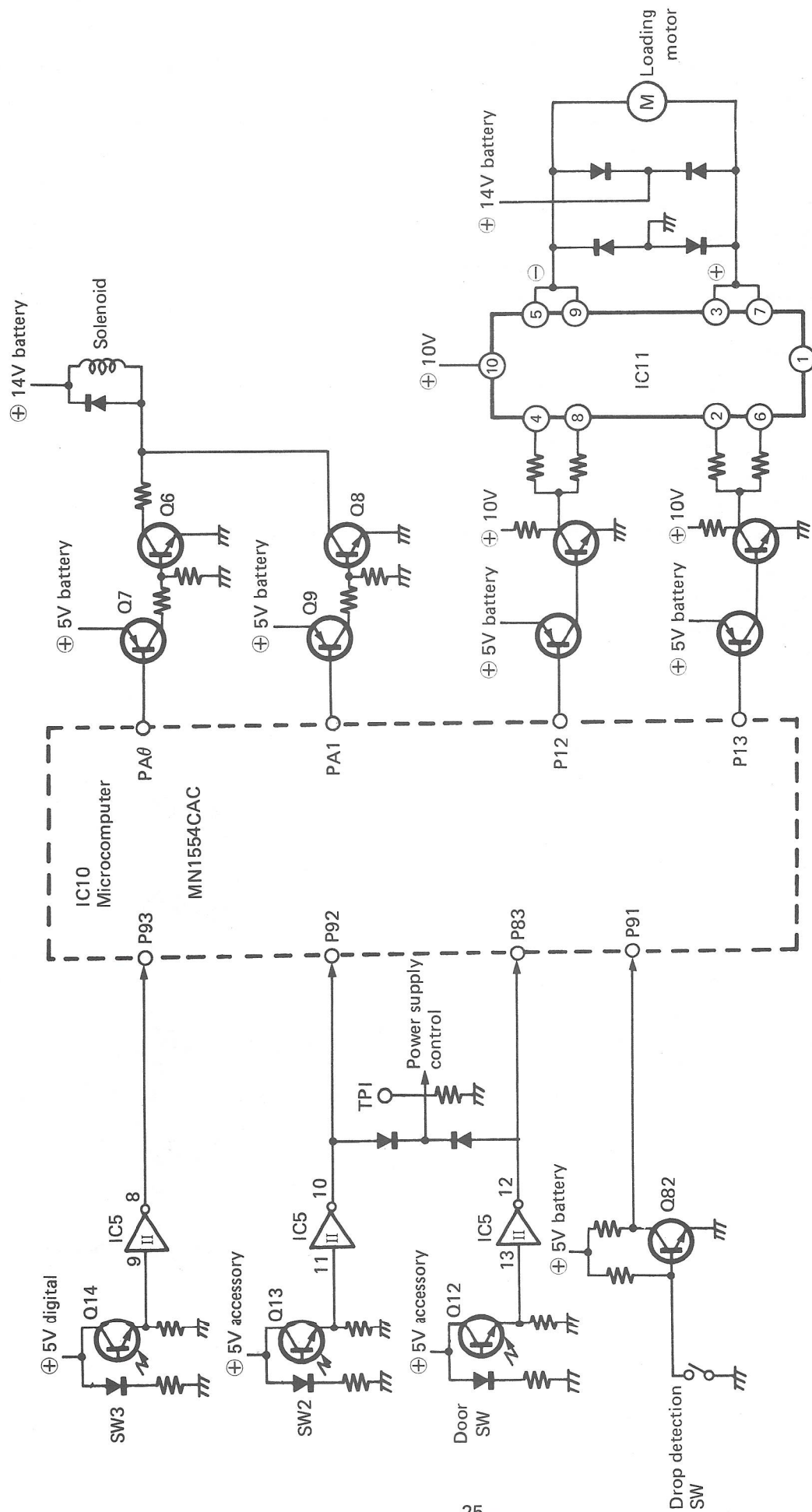


Fig. 3-2 Disk Loading Circuit

C3-3 Eject Operation

Mode	Play	Elevating	Disk pushed for ejection	Eject
Sensor	Eject Key	Door SW ON	SW3 OFF	SW2 OFF
Output	Spindle motor OFF	Solenoid ON Loading motor ON (Reverse)	Solenoid OFF Loading motor OFF	Door SW OFF
Remarks	Door opens. Elevation gear begins to rise.	Magnet plate releases. Deck is locked.	Disc moves outward.	The unloading is completed.

C3-2 Loading Operation

Mode	Disk insertion	Disk loading	Lowering	Play
Sensor	Door SW ON	SW2 ON	SW3 ON Door SW OFF	
Output	Solenoid ON Loading motor ON		Solenoid OFF	Loading motor OFF Spindle motor ON
Remarks	Door opens. Elevation gear begins to descend.	Roller gate opens. Roller turns.	Magnet plate holds disk down. Deck is released and supported by the suspension system.	The loading is completed.

3. LOADING MECHANISM and OPERATION (TYPE C)

C3-1 Disk Loading Mechanism

The following switches (sensors) detect the disk position.

- Door switch
- SW2
- SW3

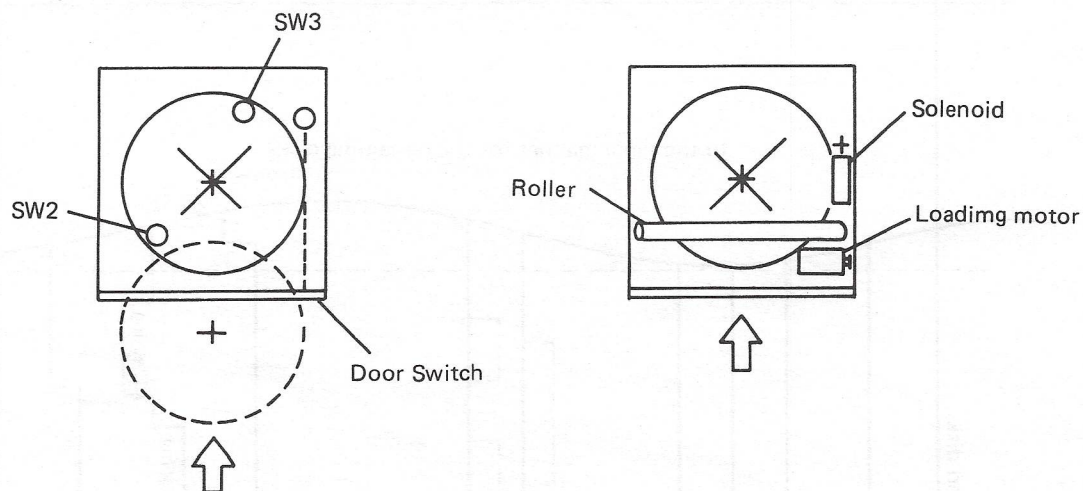


Fig. 3-1 Loading Mechanism Sensor Locations

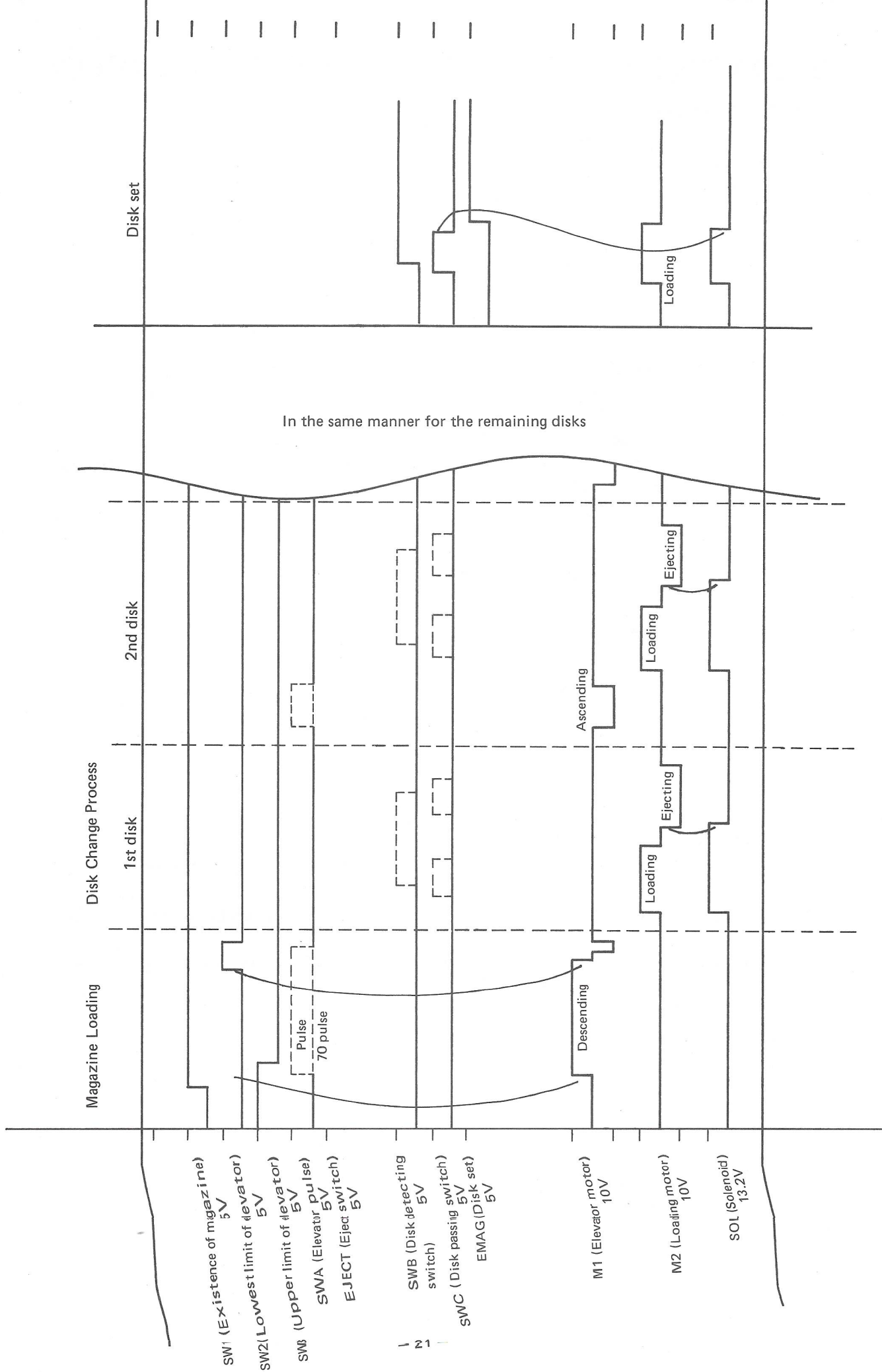
The following devices load and unload the disk:

- Loading motor : Bidirection motor to load, eject, raise, and lower the disks.
- Solenoid : Opens and closes the door, and suspension system operation.

In the play mode, the deck sit on a cushion base with the disk hold down by the magnet plate.

* The drop detection switch is a standby switch that is used to release the magnet plate by means of EJECT, should it drop when there is no disk loaded.

B3-3 CD Auto-Changer Loading Timing Chart



CD CHANGER LOADING MECHANISM (TYPE B)

B3-2 Operation of Loading Motor and Solenoid

(1) Disk loading/unloading

Disk loading to and unloading from the optical pick deck can be done by the loading motor (M2) and roller assembly.

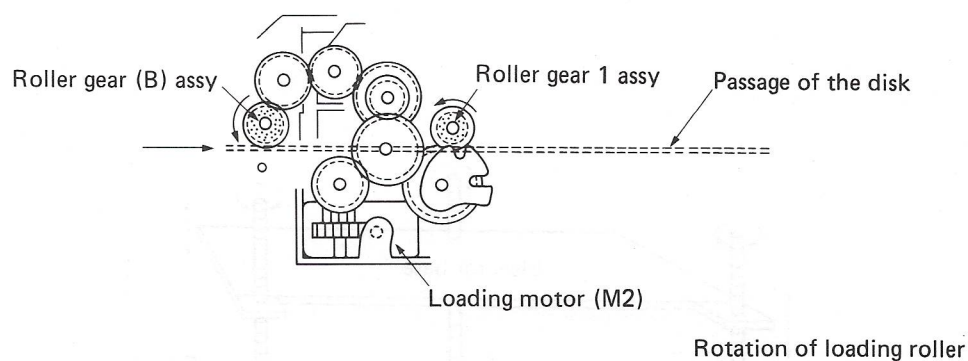


Fig. Right Side View of the Deck

(2) Unloading the disk from the magazine

When the loading motor rotates to the loading direction, the solenoid turns on, the gear lock lever goes down and engages with the cam gear. Then the cam gear pushes the disk ejecting lever once. As a result, the magazine lever is pushed, and kicks the disk out to the loading mechanism.

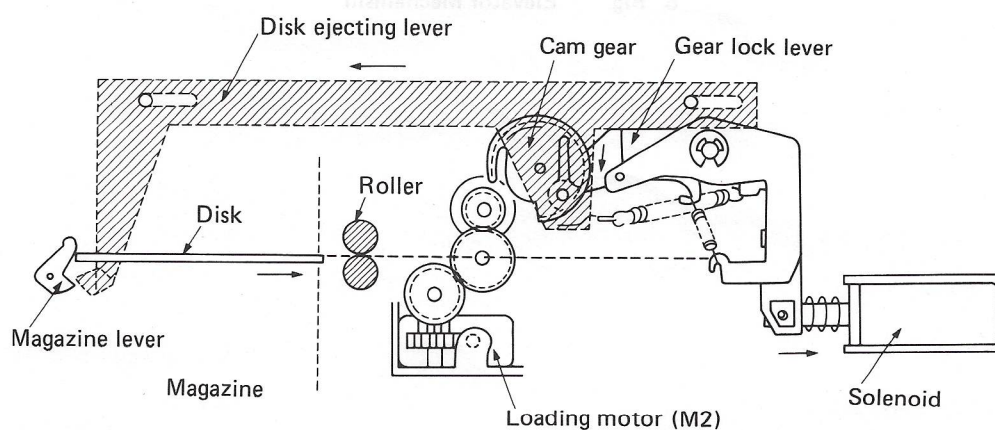


Fig. Right Side View of the Deck

After this, the loading operation is the same as Type A.

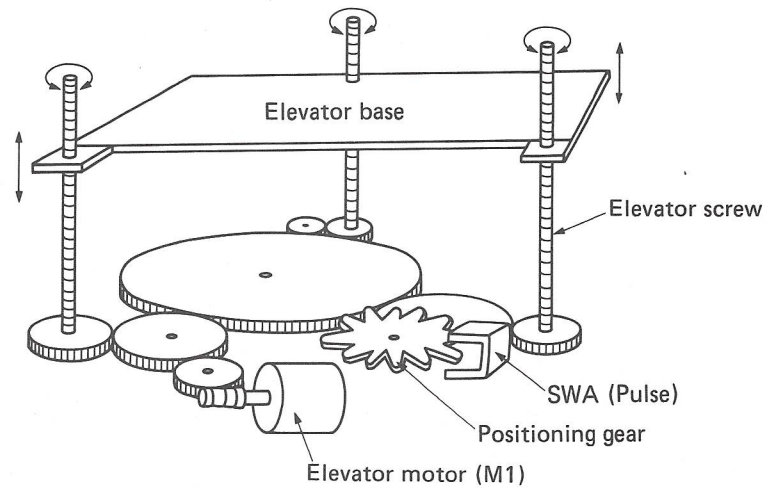
CD CHANGER LOADING MECHANISM (TYPE B)

B3-1 Elevator Motor and Disk Loading Positioning

When the magazine placed on the elevator base with cover closed, the elevator motor (M1) runs, and the elevator base moves downward until it hits SW2 and stop.

The elevating process is disk positioning.

- 1 The stop position at the SW2 is the initial position of Disc Loading.
- 2 For moving the magazine to the correct loading position, it is detected by the pulse of 70 interval.

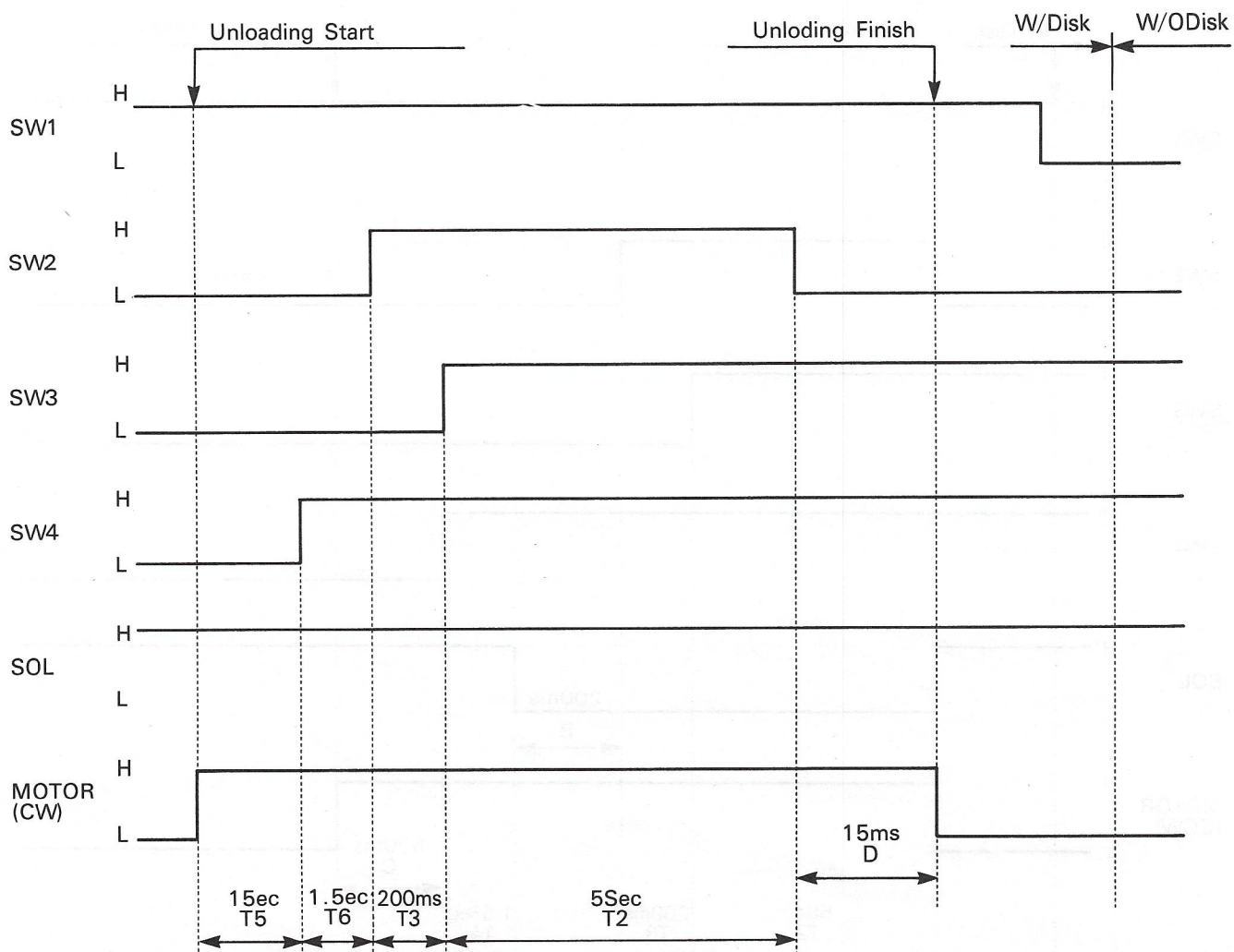


6 Fig. Elevator Mechanism

UNLOADING

W/O DUST COVER

Active H



D = 15ms

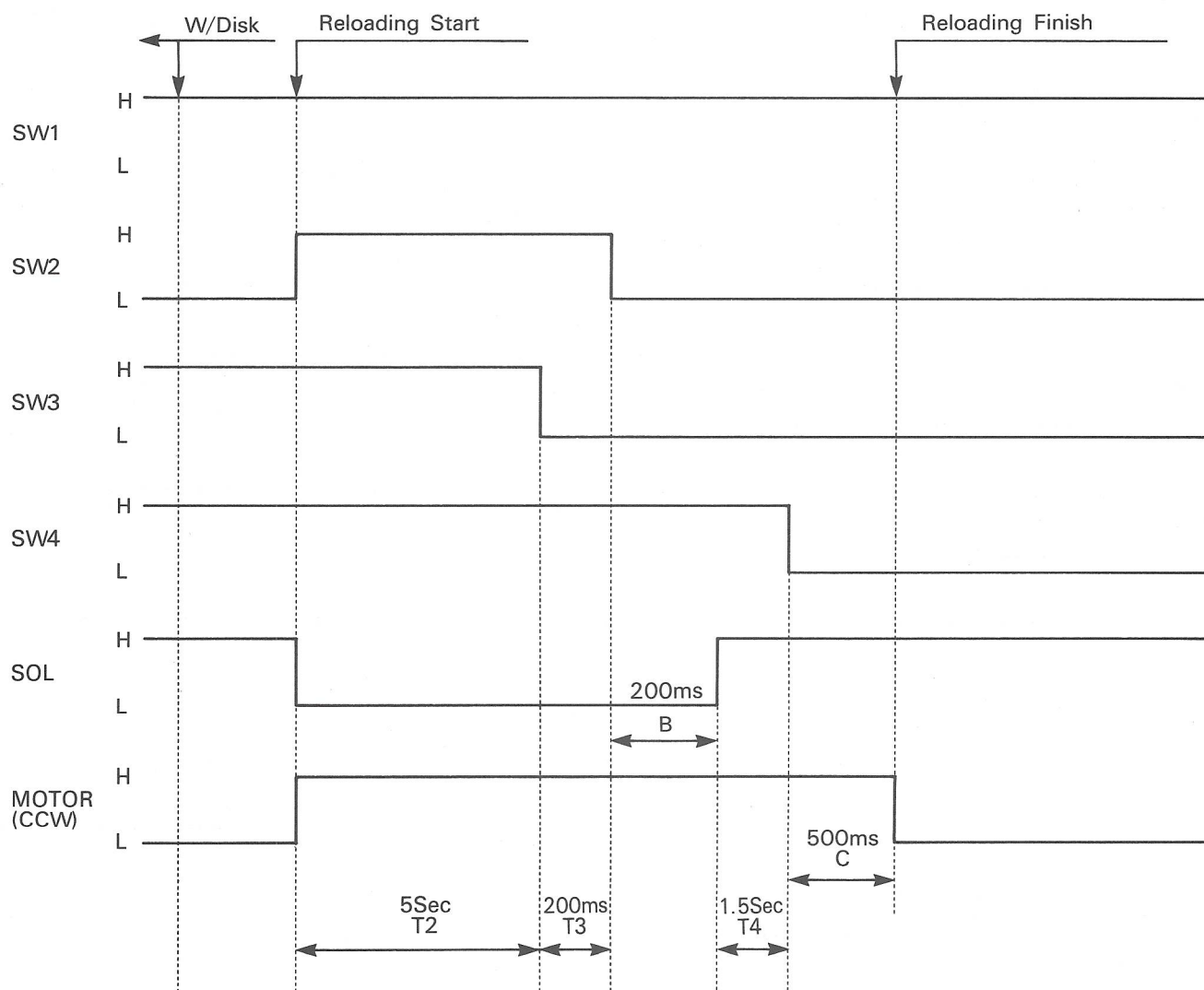
T3 = 200ms

T5 = 1Sec

T6 = 1.5Sec

RELOADING

Active H



A = 200ms

B = 200ms

C = 500ms

T2 = 5Sec

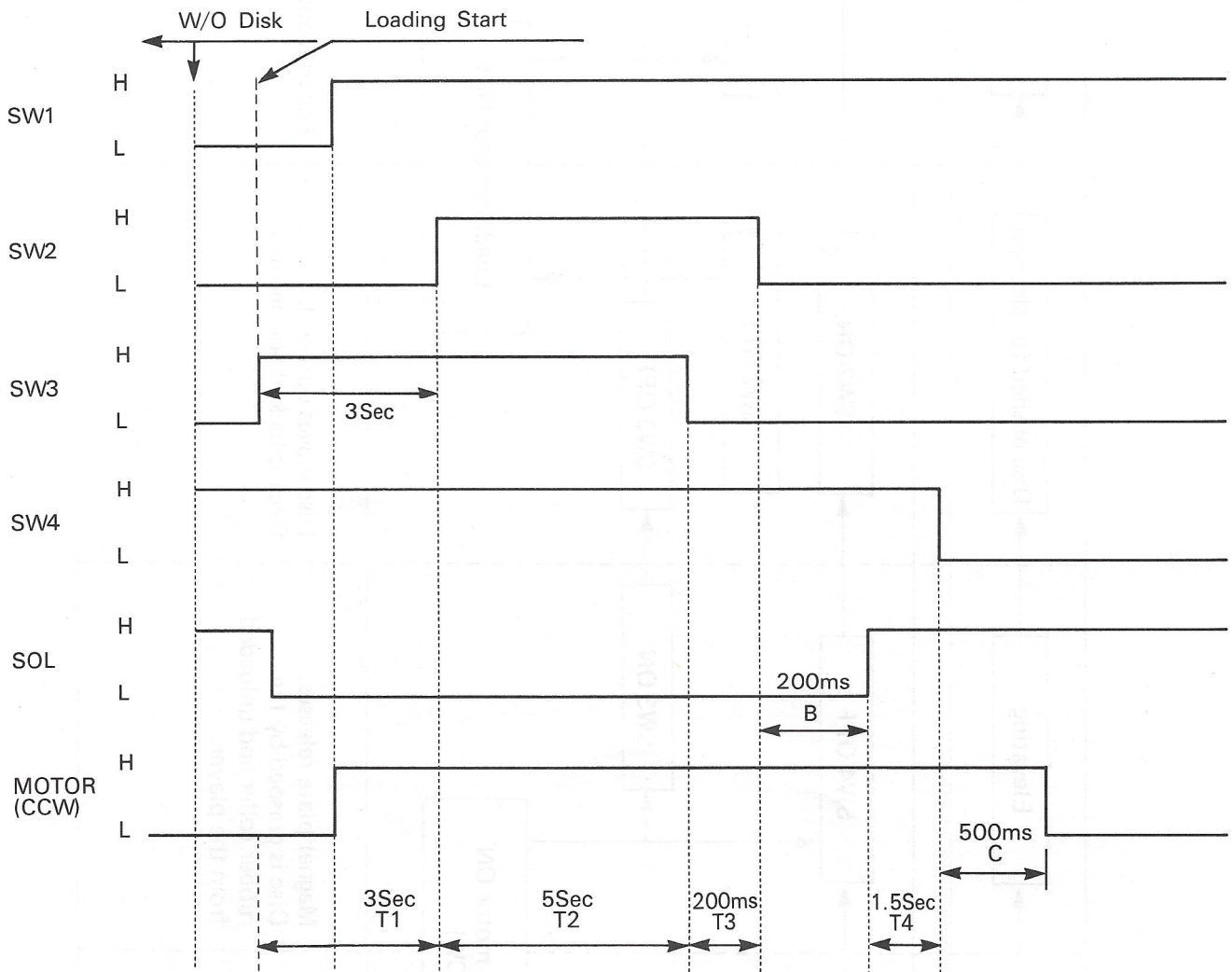
T3 = 200ms

T4 = 1.5Sec

A3-6 Timing Chart

LOADING

Active H



B = 200ms

C = 500ms

T1 = 3sec

T2 = 5sec

T3 = 200ms

T4 = 1.5sec

3-5 Eject Operation

